

Experimental VLE, consistency and fitting

After midterm · Session 2 of 6 · 180 minutes

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The question for today

Does a small fitting error mean the data and model are trustworthy?

Before class: 3-page reading · 15–20 minutes

Bring the three preparation answers. Revisit your prediction after one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion and exit question	30

The 70-minute investigation is student work with instructor support.

Activity and its reference

$$\mu_i = \mu_i^\circ + RT \ln a_i, \quad a_i = \gamma_i x_i$$

The activity is dimensionless; γ is its correction relative to an ideal liquid solution using the pure-liquid standard.

Residual: real fluid minus ideal gas at the same T, P and composition.

Excess: real mixture minus ideal solution at the same T, P and composition.

Properties → prediction → fitting

1. Verify the Antoine form, units and applicable temperature range.
2. Predict P and y from a chosen liquid model.
3. Infer activities from observations, then compare model fits.

$$\gamma_i = \frac{y_i P}{x_i P_i^{sat}(T)}$$

This inversion assumes ideal vapor and negligible Poynting correction. Convergence, consistency and prediction are separate checks.

A vapor-pressure error becomes an activity error

Synthetic point: $P=100$ kPa, $x_1=0.4$, $y_1=0.6$ and $P_1^{\text{sat}}=120$ kPa.

$$\gamma_1 = \frac{0.6(100)}{0.4(120)} = 1.25$$

Using 150 kPa instead yields $\gamma_1=1.00$. The measured mixture data did not change; the inferred liquid behavior did.

An independent check

Check one Antoine saturation pressure and one inferred activity coefficient by hand. Keep the same observations and property convention when comparing models.

One demonstration • 20 minutes

[Open the core lab](#)

Load and validate the synthetic example in Lab 08. Inspect its Antoine form and units, then fit one-parameter Margules. Compare predicted P and y with one observed point, then inspect the residual pattern.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 08, validate the supplied synthetic example and compare one-parameter Margules with NRTL at $\alpha=0.3$ on the same observations. Report one residual pattern and verify one activity coefficient by hand.

Discussion and exit question • 30 minutes

Use 20 minutes to compare results and discuss unexpected behavior.

Does a small fitting error mean the data and model are trustworthy?

Use the final 10 minutes to explain your answer individually and name one assumption that limits it.

Follow-up practice • 30–40 minutes

In Lab 08, validate the supplied synthetic example and compare one-parameter Margules with NRTL at $\alpha=0.3$ on the same observations. Report one residual pattern and verify one activity coefficient by hand.

Assessment 2 and independent checkpoint

Retain one study JSON, one comparison and a short explanation. Continue from your classroom case. The hand checkpoint is included. With pre-class reading, independent work totals 45–60 minutes.

Optional study

Choose one extension: a controlled consistency defect in Lab 03, a fixed held-out comparison, bootstrap fit spread, or your own sourced VLE data. Exact synthetic bootstrap spread does not quantify real measurement uncertainty.

[Session appendix](#) · [Module reference](#)

[Learning path](#) · [Offline package](#)

Optional work is additional. Complete the required investigation first.